Single-cell RNA sequencing reveals tumor progression and microenvironment features in immune-hot and immune-cold of hepatocellular carcinoma

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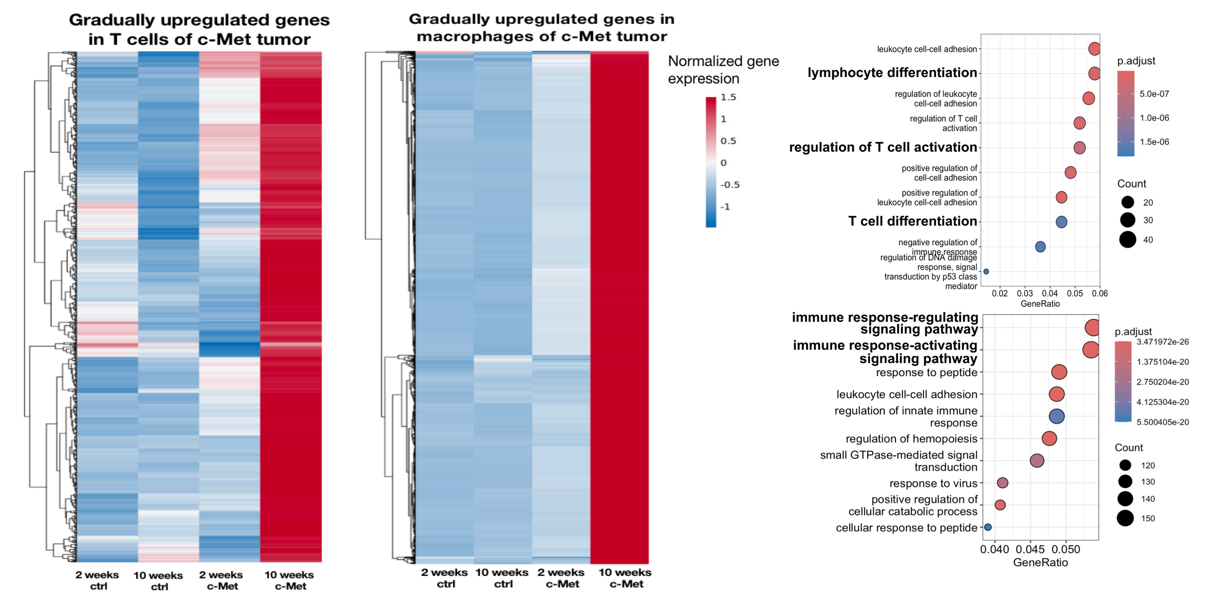
Background

Hepatocellular carcinoma (HCC) has tripled in incidence in the United States over the past 40 years, with more than 40,000 new cases diagnosed annually. While HCC can be curatively resected in the early stages, its high recurrence rate and lack of effective treatments result in a 5-year survival rate of only 18%. Tumors can be classified as hot (immunogenic) or cold (non-immunogenic) based on immune cell infiltration, with hot tumors generally being more responsive to immunotherapy and having a better prognosis. Previously, our study demonstrated unique molecular characteristics of Hispanic HCC through multi-omics analysis and identified two subtypes with different immune activity and prognosis. However, the mechanisms that drive and shape these differences in the tumor and its microenvironment remain unclear.

Methods

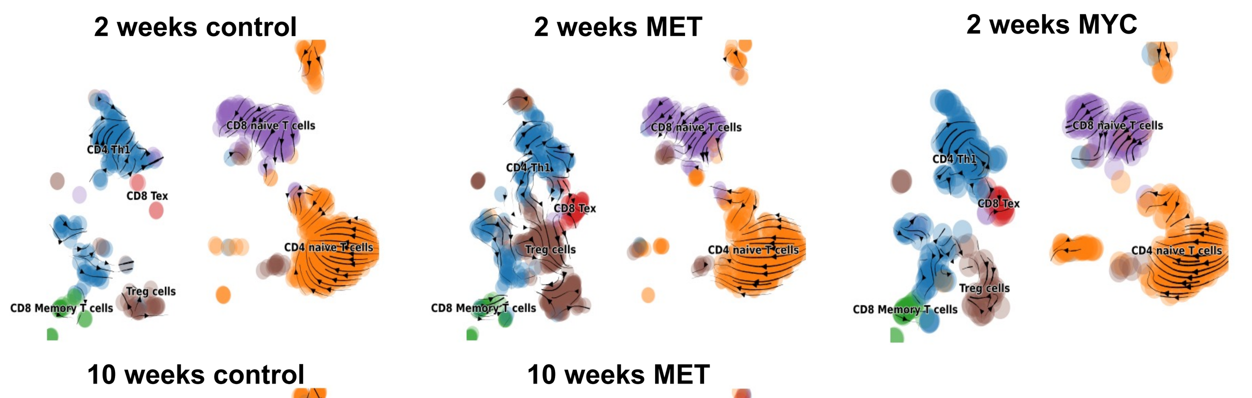
Spontaneous HCC mouse models were generated via hydrodynamic tail vein injection of plasmids encoding a mutant human β-Catenin together with either c-Myc or c-Met. The stable genomic integration of these oncogenes in hepatocytes is mediated by the Sleeping Beauty transposon system. For this study, we focused on two representative stages of tumor progression: early-stage tumors (2 weeks post-injection for both β-Catenin + c-Myc and β-Catenin + c-Met models) and late-stage tumors (5 weeks for the β-Catenin + c-Myc model and 10 weeks for the β-Catenin + c-Met model). Control groups were included at matched time points (2 replicates each). We performed single-cell RNA sequencing (scRNA-seq) on liver tissues from the two HCC tumor models and control mice across the two time points, profiling a total of 56,170 high-quality cells from 12 libraries. The scRNA-seq data were used to investigate transcriptional and functional changes across different cell types, groups, and time points. We conducted RNA velocity analysis to predict cellular trajectories and explore the functional changes of immune and non-immune cells at different time points and models. In parallel, we collected bulk RNA-seq data from isolated tumor cells and hepatocytes at corresponding time points to complement the single-cell data and investigate the molecular features of tumor cells in hot and cold tumors.

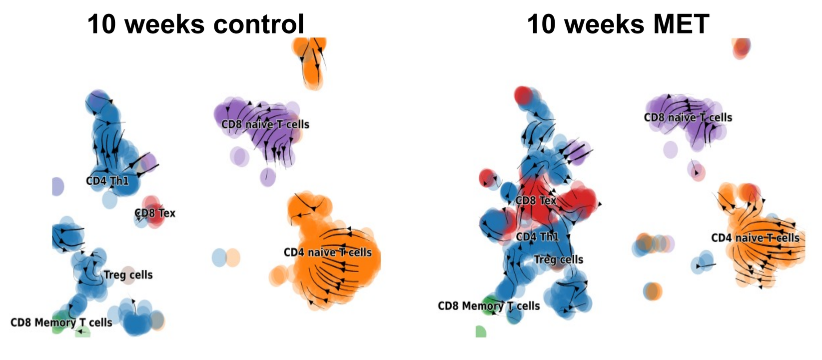
Upregulated genes in immune cells indicate enhanced immune activation in MET tumors



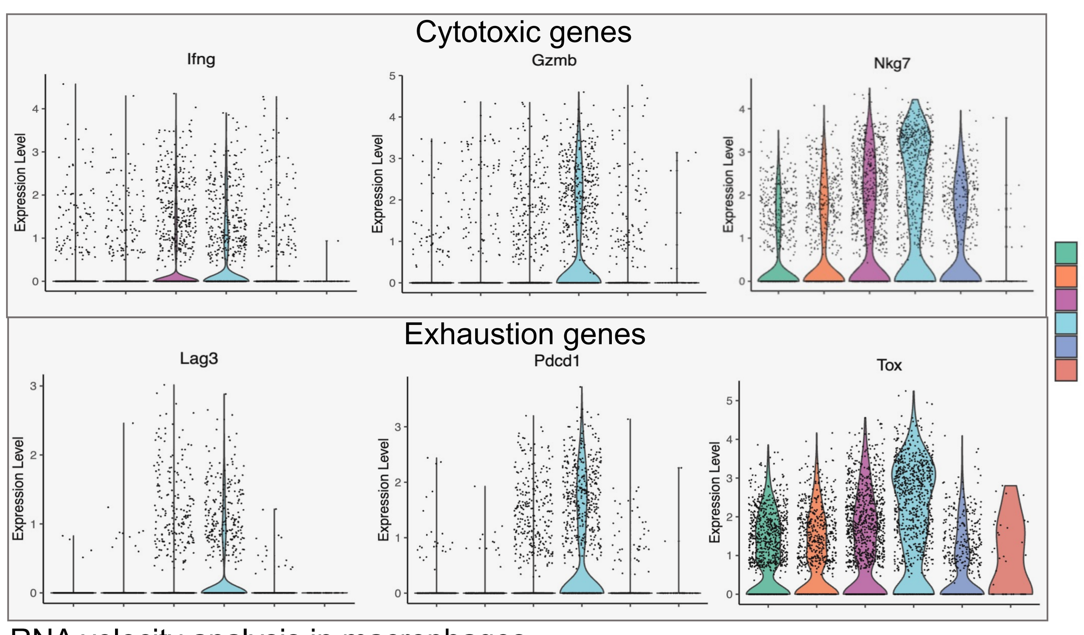
Distinct functional differentiation trajectories of T cells and macrophages revealed by RNA velocity analysis

Rapid transcriptional transitions drive T cell exhaustion in the immune hot tumor microenvironment

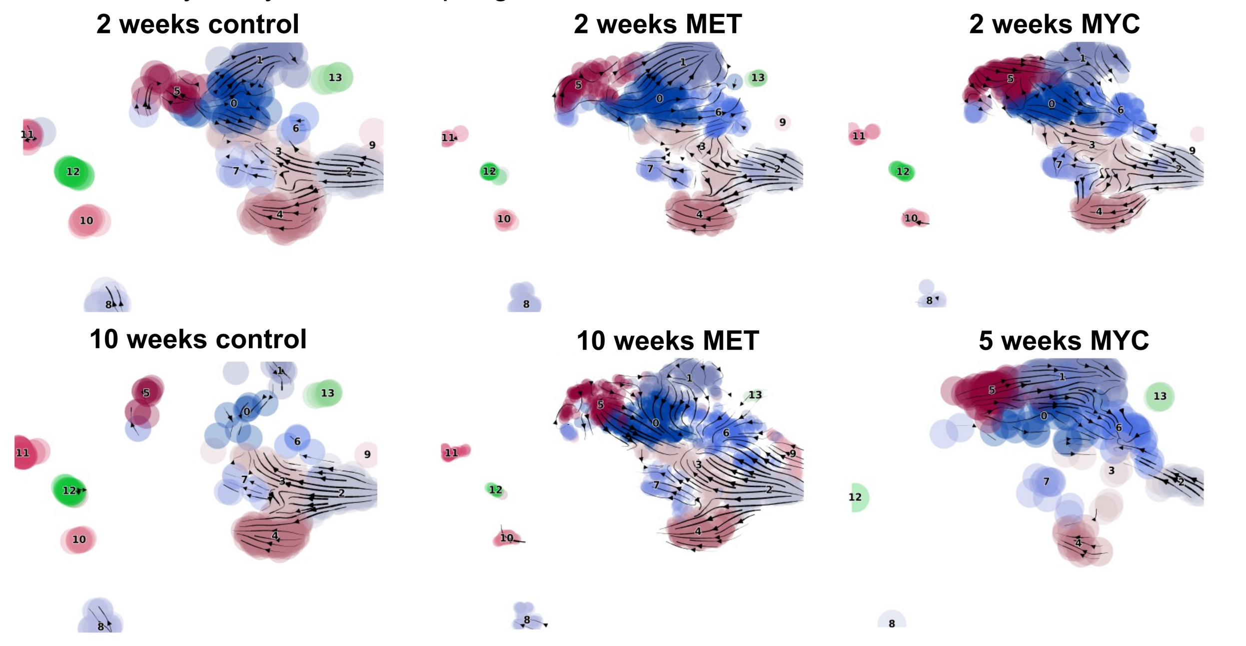




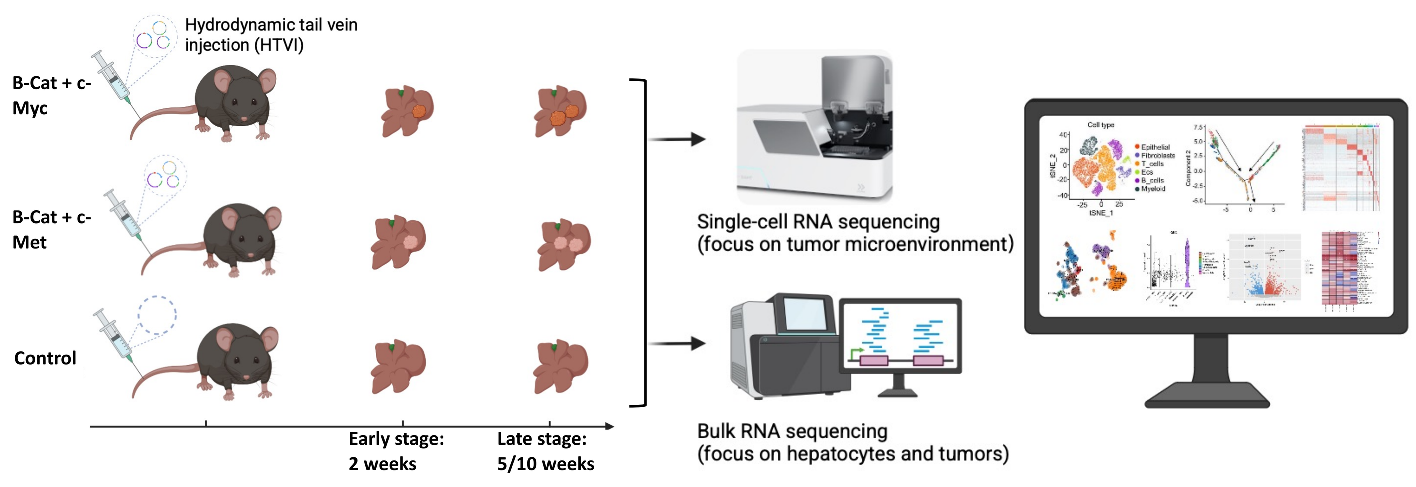
RNA velocity and marker expression patterns indicate that T cells in immune-hot tumors differentiate toward more functionally mature T cells.



RNA velocity analysis in macrophages

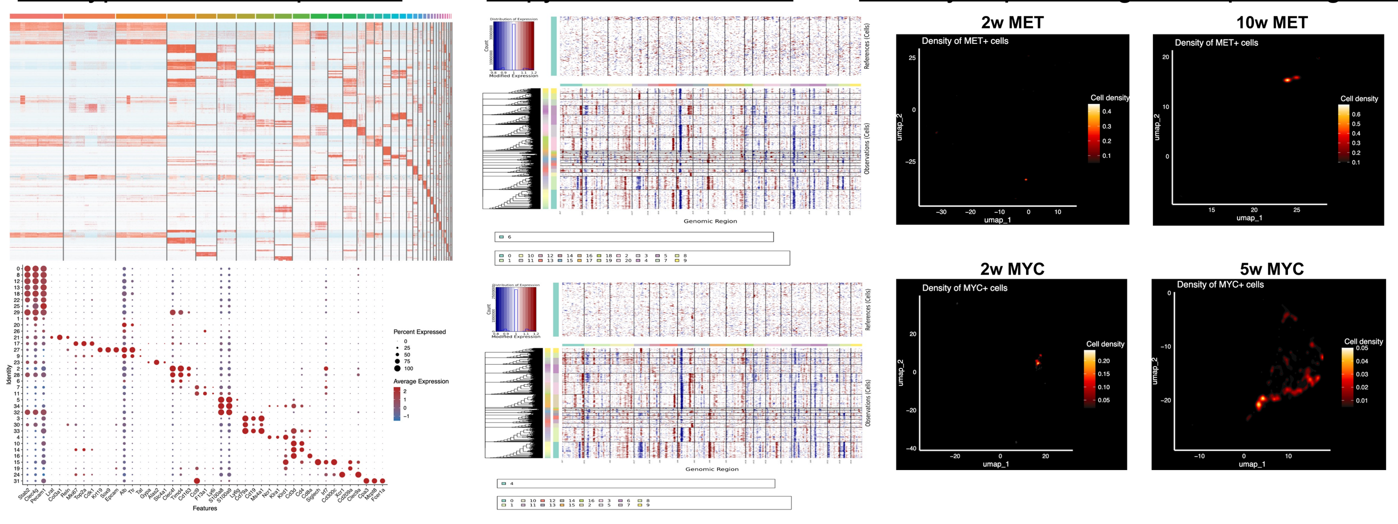


Experimental Design

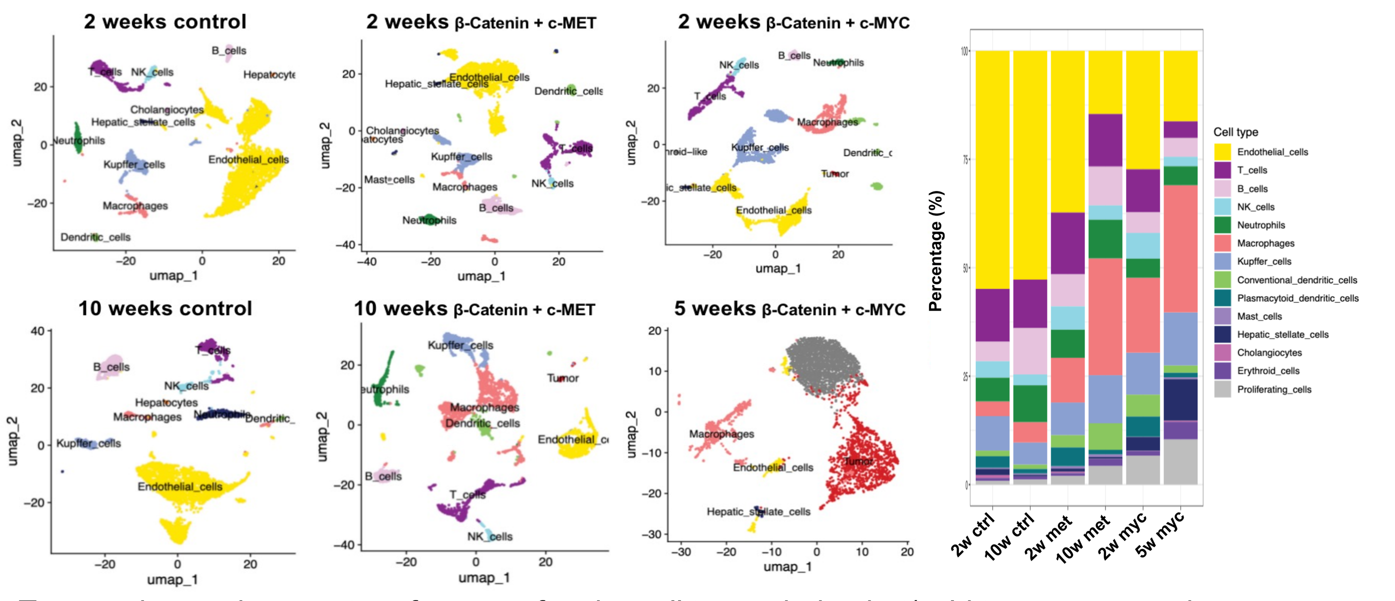


β-catenin + c-Myc and β-catenin + c-Met tumors exhibit cold and hot phenotypes

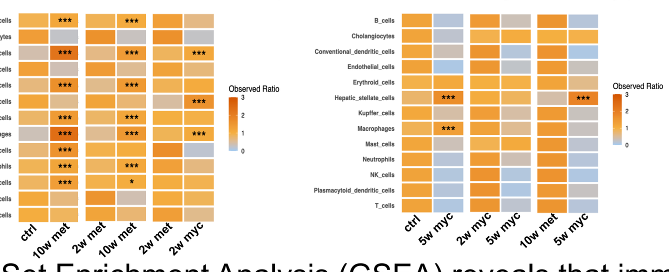
Cell type annotation process | Copy number variation | Density of plasmid gene-expressing cells



A higher diversity of immune cells was observed in the MET tumor group

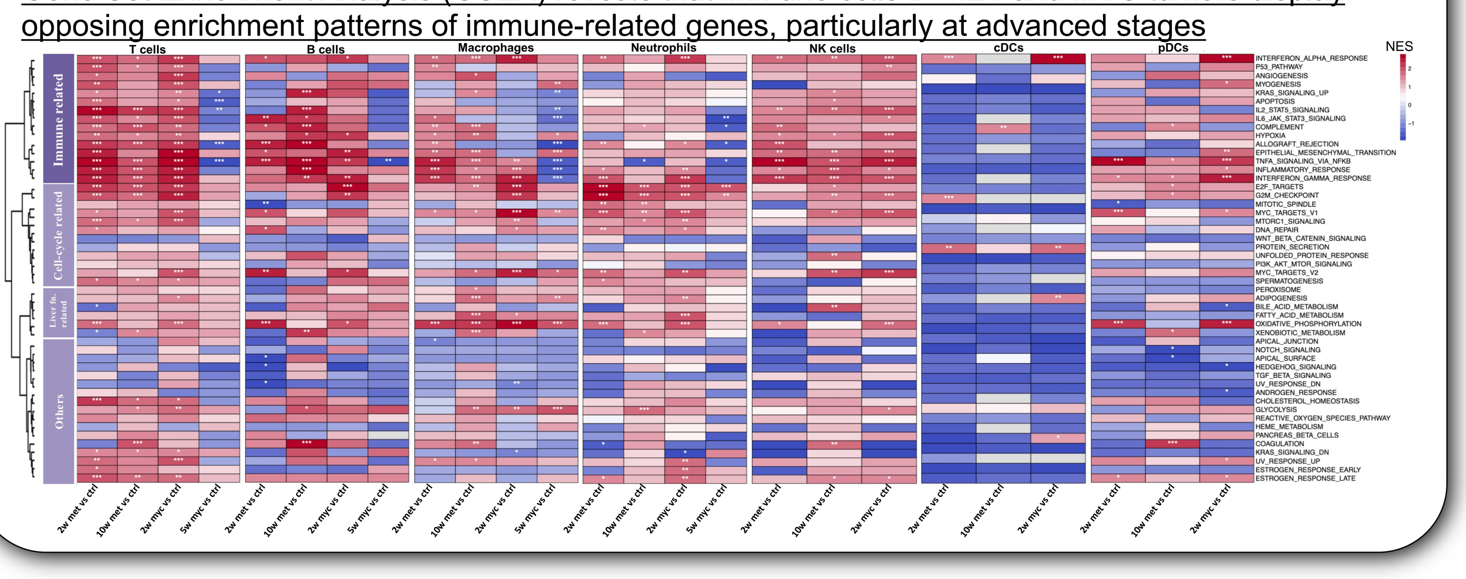


Tumor microenvironment preference of major cell types during hot/cold tumor progression



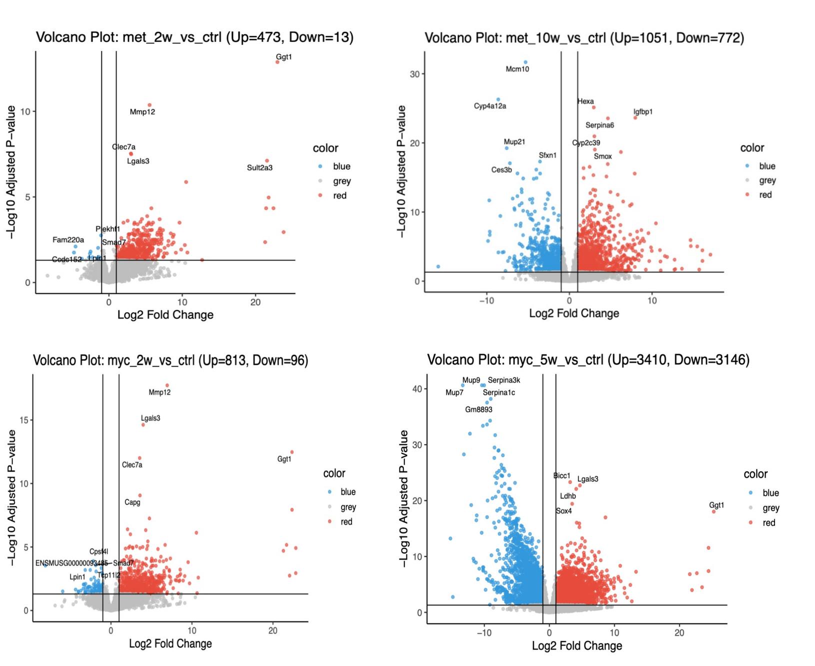
MET-driven tumors show an immune-inflammatory phenotype, with significant increases in various immune cell populations and enhanced immune function, whereas MYC-driven tumors show a non-significant decrease in the proportion of lymphocytes.

Gene Set Enrichment Analysis (GSEA) reveals that immune cells in MET and MYC tumors display opposing enrichment patterns of immune-related genes, particularly at advanced stages

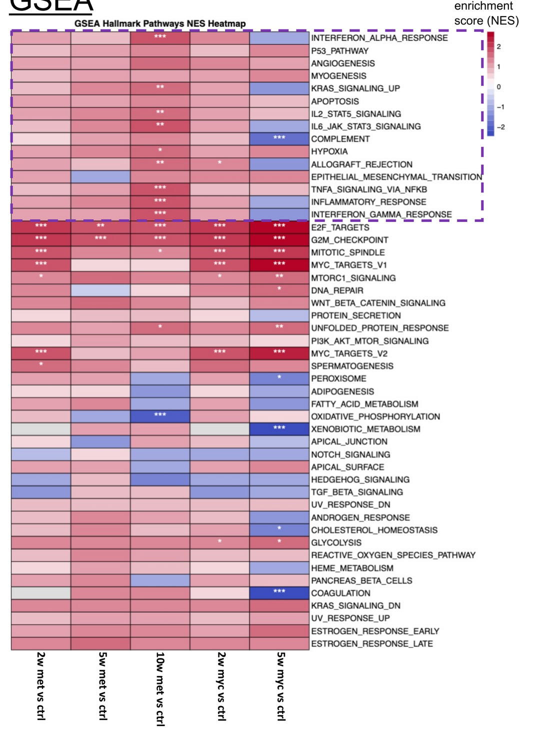


Intrinsic immune characteristics of tumor cells revealed by bulk RNA-seq data

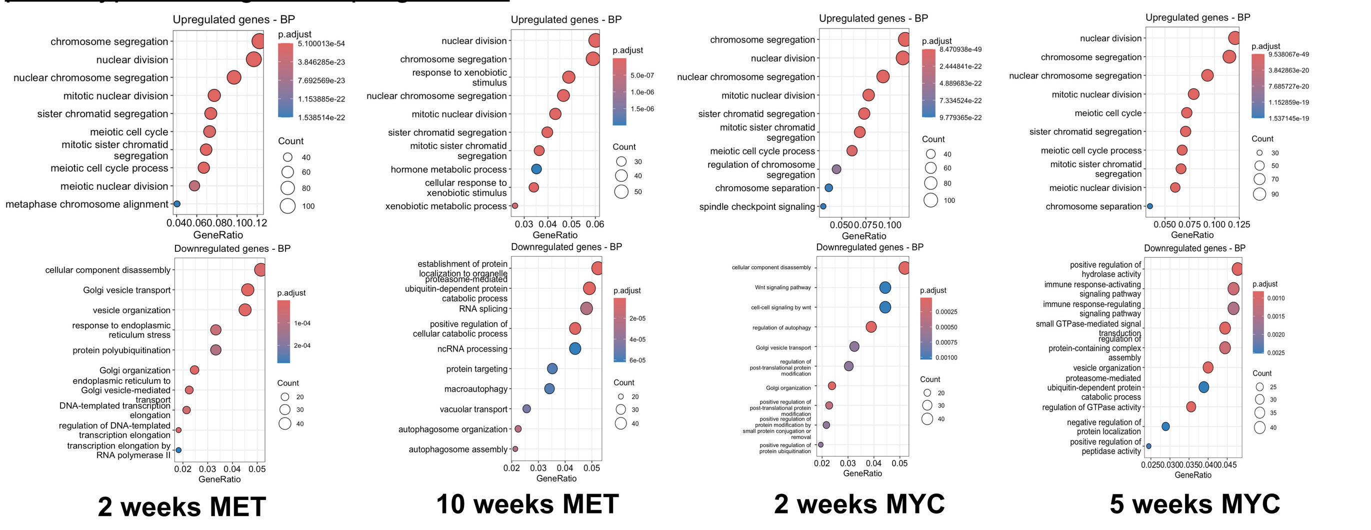
Differential Expression Analysis



GSEA



Conserved activation programs and distinct suppressed pathways drive the Hot and Cold tumor phenotypes during tumor progression



Significance:

We establish mouse models that recapitulate cold and hot HCC phenotypes observed in patients, enabling mechanistic dissection of immune state regulation. A single oncogenic partner difference reshapes not only immune cell composition but also hepatocyte-intrinsic immune programs, providing a platform to uncover drivers of tumor immune heterogeneity.

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